Best Practices for Biomedical Research Data Management

Module 9: Scientific Research Team Librarian's Role on a Research Project

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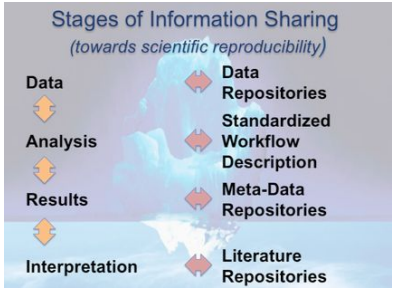

This video will highlight a librarian's role on a research team. Specifically, the video will expand on the role of an informationist for the NIH supplemental grant: Data Identification, Citation, and Reuse: A Case Study in a Neuroimaging Repository. Leah will talk about her project aims: data identification, data reproducibility, and credit for creation and authorship.

CC BY NC SA

Informationist Supplemental Grant

Dr. Jean Frazier
Dr. David Kennedy
Dr. Christian Haselgrove
Child and Adolescent NeuroDevelopment Initiative (CANDI)
UMass Med Dept. of Psychiatry

Data Identification, Citation, and Reuse: A Case Study in a Neuroimaging Repository

My Project

Three main aims of the grant:

- ➔ Data Identification
- ➔ Data Reproducibility
- ➔ Credit for Creation and Authorship

First Class Research Object

What does it mean to cite data?

Citation counts and impact factors are the currency in academic science. They are used in grant reviews, tenure decisions, and performance assessment. But this is finally starting to change:

http://www.nature.com/articles/sdata201544

Altmetric: 6 Views: 2,980 Citations: 1 More detail »

Data Descriptor | OPEN

A multimodal MRI dataset of professional chess players

Kaiming Li, Jing Jiang, Lihua Qiu, Xun Yang, Xiaoqi Huang, Su Lui & Q

Scientific Data **2**, Article number: 150044 (2015)
doi:10.1038/sdata.2015.44
Download Citation

Brain imaging
Functional magnetic resonance imaging
Magnetic resonance imaging
Problem solving

Received: 22 J
Accepted: 16 J
Published onli

Data Records

This dataset is publicly available at Li *et al.*, 2015 (Data Citation 1: FCP/INDI doi:10.15387/fcp_indi.pro.wchsuchess). The MRI and QA data can be accessed at http://fcon_1000.projects.nitrc.org/indi/pro/wchsu_li_index.html. The total size of this dataset is approximately 60GB. It contains raw MRI data, QA reports, and QA extras. These data were separately packaged and compressed so that users can download necessary packages in a multi-thread way. The contents and data structures of these packages are detailed as follows.

Mixed Datasets

In neuroimaging, the practice of posting datasets to shared, open access repositories is well established. However, reusing data, especially creating new datasets from multiple sources presents a problem for citing and maintaining the chain of credit.

➔ Neuroimaging Informatics Tools and Resources Clearinghouse (NITRC)

NITRC
The source for neuroinformatics tools & resources
Neuroimaging data repository
Cloud computing environment

SEARCH Tools/Resources GO
Member login | Register | Help
Select Language
Powered by Google Translate

Browse by domain
Browse by software
Browse by data resources
Browse by diagnosis

Find neuroimaging tools here:

Examples:
- *modeling OR simulation*
- *morphology AND animation*
- *segmentation NOT Linux*
- *"region of interest"*

SEARCH
Search Builder

Featured tool/resource:

ENIGMA-Viewer: Visualizing Effect Sizes in Meta-Analysis

ENIGMA-Viewer is an interactive visualization tool for neuroscientists to examine DTI data and compare effective sizes and associated genetics information in meta-analysis. The program supports the visualization of sub-cortical and cortical regions. The user may upload your own data for comparison.

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Latest News

[NITRC Community - May 30 - 1 comment]
ABIDE Preprocessed Data Mirrored in NITRC-IR
We are pleased to announce that the ABIDE Preprocessed data is now available for download from NITRC-IR. This includes data preprocessed by CCS, C-PAC, DPARSF, and NIAK as well as FreeSurfer, CIVET, and ANTS runs and two types of QA. More information...

[NITRC Community - May 18 - no comments]
NITRC at SOBP 2017
Visit the NITRC table in the Sapphire Foyer at Society of Biological Psychiatry meeting in San Diego. Say Hello, Register a Resource, Find some Data...!

[NITRC Community - May 5 - no comments]
NITRC restoration and new redundancy plans
NITRC experienced a site-wide outage on April 20th. A failure in a redundant cooling system caused a set of hardware failures which took the data storage array offline. While our hosts worked on the hardware restoration, the NITRC team decided to create...

[NITRC Community - Apr 14 - no comments]
NITRC Image Repository release
We are pleased to announce the release of an updated version of the NITRC Image Repository (NITRC-IR). This release updates NITRC-IR to XNAT 1.7, modernizing the look and functionality of the repository. See the new NITRC-IR at http://www.nitrc.org/ir/. Please...

[MRTool - Apr 10 - no comments]
MRTool v1.4.1 * NEW RELEASE ***
A new version of MRTool has been just released. Download it and check the new features...

[MRTool - Apr 3 - no comments]
MRTool at OHBM Vancouver
This year MRTool goes to Canada! Visit us at poster 4134 and 3782 ... 1) Poster Number: 4134 Title: MR Tool: a SPM-based toolbox for structural MR imaging analyses Category: Modeling and Analysis Methods / Methods Development 2) Poster Number:...

[The Grid Code Analysis Toolbox - GridCAT - Mar 28 - no comments]
GridCAT v1.03 released
Version 1.03 of the GridCAT does not use the iscomplex function anymore. This function is normally delivered with SPM, but this dependency has caused an issue.

more news

Community

- General community forum
- Funding opportunities
- Sustaining NITRC
- Submit community news
- Submit tool/resource
- Publications
- Conferences and workshops
- Career opportunities

[20,153 registered users]

Recently active forums

more forums

SPM for fNIRS toolbox: help
RE: error during temporal analysis
[67 posts, last post 46 minutes ago]

CONN : functional connectivity toolbox: help
RE: ICA images
[5829 posts, last post 2 hours ago]

BrainNet Viewer: help
RE: Problem Saving Images
[356 posts, last post 21 hours ago]

Recently updated files

more updates

DWI/DTI Quality Control Tool: DTIPrep
DTIPrep: DTIPrep1.2.8_MacOSX
Latest file: **May 31**

PVS_Enhance
pvs_enhance: PVS_Enhance_ver1.0
Latest file: **May 30**

MRIcroGL
mricrogl: MRIcroGL 28-May-2017 with dcm2niix
Latest file: **May 28**

Recently registered

more tools/resources

Jun 3 calcFD - Calculate the fractal dimensionality of a 3D structure
May 29 iElectrodes
Arterial Territories of the Neonatal Brain: automated classification
May 28 PVS_Enhance

[943 registered tools/resources]

Our attempt at a solution:

http://doi.virtualbrain.org/

Logged in as: Guest | Login | Register

doi:

Search | Advanced

Home | Tools

- Projects
  - Recent
  - Favorite
  - My projects
  - Other projects
- Stored Searches
- Data

candidoi currently contains 3 projects, 121 subjects, and 121 imaging sessions.

Projects | Subjects | MR | PET | CT

ID | Name | Description

Keywords | Investigator (SELECT)

Submit

Projects

Internet Brain Segmentation Repository
Project ID: ibsr
This is a **public** project.
Request write access to this project.

CANDI Share Schizophrenia Bulletin 2008
Project ID: cs_schizbull08
Version 1.1 of the CANDI Share Schizophrenia Bulletin 2008 data.
This is a **public** project.
Request write access to this project.

Recent Data Activity

CANDI | University of Massachusetts Medical School UMASS | powered by XNAT

Key Features

DOI identifiers resolving to informational landing pages

DataCite Metadata Schema (V 3.1)

Three levels of DOI - image, project, functional - allows for identification of mixed sets, their components, and sources

Relational tags (IsPartOf, IsCitedBy, HasSource) iterate citations to both past and future data usage

Continuing Considerations

Rights and access issues

Discoverability - citation indexes?

Institutional buy-in

Scalability